

Material Safety Data Sheet

SECTION 1 IDENTIFICATION

Product Name: SPIRODICLOFEN 24 PERCENT W/V SC

Chemical name: 3-(2, 4-dichlorophenyl)-2-oxo-1-oxaspiro[4.5]dec-3-en-4-yl 2,2-dimethylbutanoate

Other Means of identification: /

Recommended use of the chemical and restrictions on use: This material is a kind of pesticide.

Company / Undertaking Identification

Company Name: Shandong Kangqiao Bio-technology Co., Ltd.

Address: Lvyi Industrial Park, 256500, Boxing County, Shandong Province, China

SECTION 2 HAZARDS IDENTIFICATION

Classification of the substance or mixture:

Sensitisation, skin Category 1

Hazardous to the aquatic environment, long-term hazard Category 4

GHS Label elements, including precautionary statements:



Signal word: Warning

Hazard statement(s): May cause an allergic skin reaction. May cause long lasting harmful effects to aquatic life.

Precautionary statement(s):

Prevention:

Avoid breathing dust / fume / gas / mist / vapors / spray. Contaminated work clothing should not be allowed out of the workplace. Wear protective gloves. Avoid release to the environment.

Response:

If on skin: Wash with plenty of water. If skin irritation or rash occurs: Get medical advice/attention.

Specific treatment (see safety data sheet). Take off contaminated clothing and wash it before reuse.

Storage: /

Disposal: Dispose of contents/container in accordance with local / regional / national / international regulations.

Other hazards which do not result in classification: /

SECTION 3 COMPOSITION/INFORMATION ON INGREDIENTS

Component	CAS No.	Concentration, w/v, g/L
Spirodiclofen	148477-71-8	240
Adjuvants	/	760

SECTION 4 FIRST AID MEASURES

Description of necessary first aid measures

If inhaled: If breathed in, move person into fresh air. If not breathing, give artificial respiration. Consult a physician.

In case of skin contact: Wash off with soap and plenty of water. Consult a physician.

In case of eye contact: Rinse thoroughly with plenty of water for at least 15 minutes and consult a physician.

If Ingestion: Rinse mouth with water. Consult a physician.

Most important symptoms/effects, acute and delayed: /

Indication of immediate medical attention and special treatment needed, if necessary: /

SECTION 5 FIREFIGHTING MEASURES

Suitable extinguishing media: Use foam, chemical powder, carbon dioxide or water spray.

Special hazards arising from the chemical: This material may decompose and burn at high temperature and fire and release toxic fumes.

Special protective actions for fire-fighters: Wear self-contained breathing apparatus for firefighting if necessary. Use water spray to cool unopened containers.

SECTION 6 ACCIDENTAL RELEASE MEASURES

Personal precautions, protective equipment and emergency procedures: Use personal protective equipment. Avoid breathing vapors, mist or gas. Ensure adequate ventilation. Remove all sources of ignition. Evacuate personnel to safe areas.

Environmental precautions: Prevent further leakage or spillage if safe to do so. Do not let product enter drains.

Methods and materials for containment and cleaning up: Contain spillage and then collect with pump and place in a clean container for disposal according to local regulations.

SECTION 7 HANDLING AND STORAGE

Precautions for safe handling: Wear protective gloves / eye protection / face protection / protective clothing. Avoid contact with skin and eyes. Avoid inhalation of vapour or mist. Keep away from sources of ignition - No smoking.

Conditions for safe storage, including any incompatibilities: Store in cool place. Keep container tightly closed in a dry and well-ventilated place. Containers which are opened must be **carefully** resealed and kept upright to prevent leakage. Keep away from flammable materials and oxidizers.

SECTION 8 EXPOSURE CONTROLS/PERSONAL PROTECTION

Control parameters: /

Appropriate engineering controls: Local exhaust ventilation or a process enclosure ventilation system may be required.

Individual protection measures

Eye/face protection: Safety glasses with side shields. Chemical goggles. Contact lenses may pose a special hazard; soft contact lenses may absorb and concentrate irritants.

Skin protection: Wear chemical protective gloves, e.g. PVC. Wear safety footwear or safety gumboots, e.g. Rubber. Impervious clothing.

Respiratory protection: Selection of the Class and Type of respirator will depend upon the level of breathing zone contaminant and the chemical nature of the contaminant.

Thermal hazards: /

SECTION 9 PHYSICAL AND CHEMICAL PROPERTIES

Appearance (physical state, colour etc)	Off-white viscous liquid.
Odour	Weak characteristic odour
pH	5.0-8.0
Wet Sieve Test	Min. 98% pass 75µm test sieve
Persistent Foam	25mL Max. after 1 minute
Suspensibility	90% Min.
Porability	Residue after Pouring: 5.0% Max. Residue after Rinsing: 0.5% Max.
Melting point/freezing point	101.6 °C
Initial boiling point and boiling range	N/A
Flash point	Non-flammable
Evaporation rate	N/A
Flammability (solid, gas)	N/A
Upper/lower flammability or explosive limits	N/A
Vapour pressure	N/A
Vapour density	N/A
Relative density	1.07 ± 0.05 g/mL at 20°C
Solubility(ies)	Soluble in water.
Partition coefficient: n-octanol/water	N/A
Auto-ignition temperature	N/A
Decomposition temperature	N/A
Viscosity	380.0 ± 2.0mPa·S

SECTION 10 STABILITY AND REACTIVITY

Reactivity: /

Chemical stability: This material is stable in normal temperature.

Possibility of hazardous reactions: /

Conditions to avoid: Spark, static electricity and high temperature.

Incompatible materials: Flammable materials and oxidizers.

Hazardous decomposition products: Oxycarbides, chlorides, etc.

SECTION 11 TOXICOLOGICAL INFORMATION

Information on the likely routes of exposure: Inhaled, swallowed, skin, eyes.

Symptoms related to the physical, chemical and toxicological characteristics: /

Acute health effects: Accidental ingestion of the material may be harmful and cause cough and throat pain. Oral intake may cause headache, giddiness, vomit and other symptoms. This material may produce skin and eyes irritation.

Chronic health effects: Repeated or prolonged exposure to this material may cause skin allergy.

Numerical measures of toxicity(such as acute toxicity estimates):

Acute Oral Toxicity (Rat) : $LD_{50} = 5000 \text{ mg/kg}$, Be classified in GHS Category 5 or unclassified.

Acute Dermal Toxicity (Rat) : $LD_{50} > 5000 \text{ mg/kg}$, Be classified in GHS Category 5.

Acute Inhalation Toxicity (Rat) : $LC_{50}(4h) > 4.12 \text{ mg/L}$, Could not be classified according to GHS.

Acute Eyes Irritation (Rabbit) : Non-irritant, Could not be classified according to GHS

Acute Dermal Irritation (Rabbit) : Non-irritant, Could not be classified according to GHS

Skin Sensitization(Guinea Pig) : Non-sensitiser

ADI : 0.01 mg/kg bw

SECTION 12 ECOLOGICAL INFORMATION

Toxicity to Fish :	Acute toxicity for Zebra fish $LC_{50}(96h) = 15.6 \text{ mg a.i/L}$
Toxicity to Bird:	Acute oral toxicity for Quail $LD_{50}(7d) > 1000 \text{ mg a.i/L bw}$
Toxicity to Bee :	Acute oral toxicity $LC_{50}(48h) > 2000 \text{ mg a.i/L}$; Acute dermal toxicity $LD_{50}(48h) > 100 \mu\text{g a.i./bee}$
Toxicity to Daphnia Magna:	$LC_{50}(48h) = 3.91 \text{ mg a.i./L}$
Toxicity to Green alga:	$EC_{50}(72h) = 4.09 \times 10^{-2} \text{ mg a.i/L}$
Toxicity to Trichogramma:	security coefficient > 11.1
Toxicity to Silkworm:	$LC_{50}(96h) = 2000 \text{ mg a.i/L folium mori}$
Toxicity to Earthworm:	$LC_{50}(14d) > 100 \text{ mg a.i./kg dry soil}$
Toxicity: May cause long lasting harmful effects to aquatic life.	
Persistence and degradability: /	
Bioaccumulative potential: /	
Mobility in soil: /	
Other adverse effects: /	

SECTION 13 DISPOSAL CONSIDERATIONS

Disposal methods: Burial in a land-fill specifically licensed to accept chemical. Reuse of broken container is forbidden.

SECTION 14 TRANSPORT INFORMATION

UN Number: 3082.

UN proper shipping name: ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S.

Transport hazard class(es) : 9.

Packing group, if applicable: III.

Environmental hazards: /

Special precautions for user: /

SECTION 15 REGULATORY INFORMATION

Regulations: This Material Safety Data Sheet is in compliance with the following national standards: GB 16483-2008, GB 13690-2009, GB/T 15098-2008, GB 18218-2009, GB 15258-2009, GB 6944-2012, GB 190-2009, GB 191-2009, GB 12268-2008, GA 57-1993, GBZ 2-2007 as well as the following national regulations: Dangerous Goods Transport Administrative Regulation [Published by the Ministry of Railways, 2008], Dangerous Chemicals Safety Administrative Regulation [Published by the State Council, 2011].

SECTION 16 OTHER INFORMATION

References	UN Recommendations on the Transport of Dangerous Goods Model Regulations UN Globally Harmonized System of Classification and Labeling of Chemicals
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Note 1: When products contain two or more hazardous substances, Safety Data Sheets should be prepared based on the risk of the mixture.

Note 2: Manufacturer / supplier should ensure the correctness of the information contained in the safety data sheets, and updated in a timely manner.

Note 3: As a result of product features without the existence of certain information or no data available (such as boiling point does not exist for the solid) in the table with "/" logo.

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